

EXHIBIT I

**IN THE UNITED STATES DISTRICT COURT FOR THE
DISTRICT OF COLORADO**

United States of America

v.

**CEMEX, Inc.
Lyons Cement Plant
Lyons, Colorado**

Civil Action No. 09-cv-00019-MSK-MEH
U.S. District Court for the District of Colorado

**EXPERT REBUTTAL REPORT
OF
WALTER L. GREER**

A handwritten signature in cursive script, reading "Walter L. Greer", is written over a horizontal line.

June 24, 2011

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EXECUTIVE SUMMARY

1. INTRODUCTION

A. Information Required by Federal Rules of Civil Procedure

This Rebuttal Report contains my opinions, conclusions, and the reasons therefore.

1. Appendix A to the Expert Report of Walter L. Greer, March 1, 2011, contains a statement of my qualifications.
2. Appendix A affixed hereto lists the references, data, and other information I considered in forming the opinions I express in this Rebuttal Report.
3. Appendix C to the Expert Report of Walter L. Greer, March 1, 2011, lists the publications that I have authored in the past ten years.
4. The compensation I received for the preparation of this Rebuttal Report is included at the end of my statement of my qualifications in Appendix A of the Expert Report of Walter L. Greer, March 1, 2011.
5. I have never before made any testimony as an expert at a trial or by deposition.
6. The opinions expressed in this Rebuttal Report are my own and are based on the data and facts available to me at the time of writing. Should additional relevant or pertinent information become available, I reserve the right to supplement the discussion and findings in my report.

B. Areas of Requested Opinion

I have been asked by the United States Department of Justice (DOJ) to review the reports of Messrs. John E. Hofmann,¹ Gerald L. Young,² and Dr. Kevin Davis³ tendered on behalf of CEMEX, Inc. (CEMEX) in this matter on May 10, 2011 and to identify and rebut certain assertions made by those gentlemen for which I have an alternative view.

C. Experience

Please see the Expert Report of Walter L. Greer, March 1, 2011, submitted in this matter that sets forth my qualifications and experience.

¹ Expert Report of John E. Hofmann, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado; John E. Hofmann, President and CEO of Trinity Consultants, Dallas, Texas; May 10, 2011 (hereinafter "Hofmann").

² Expert Report of Gerald L. Young, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, May 10, 2011 (hereinafter "Young").

³ Expert Report of Kevin Davis, PhD, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, May 10, 2011 (hereinafter "Davis").

2. MARCH 1997 CALCINER MODIFICATION

Mr. Hofmann and Mr. Young contend that the 1997 modification to the internal geometry of CEMEX's calciner combustion chamber improved the efficiency of the pyroprocessing system and required less coal to be burned in the kiln, thereby resulting in less NOx emissions from both the calciner and kiln. They base this contention in large part on the proposition that modification to the combustion chamber resulted in: (a) an increase in the static pressure drop, which supposedly reduced gas flow throughout the entire pyroprocessing system; (b) a change in the fuel consumption balance (split) between the calciner and kiln; and (c) a change in the degree of calcination of feed exiting the calciner and entering the kiln. As discussed below, these individual propositions are not correct. This undermines the experts' conclusion that this modification resulted in an overall decrease in fuel NOx emissions as well as a decrease in thermal NOx emissions from the kiln.

A. Modification of the Calciner in 1997 Did Not Significantly Impact the Static Pressure Drop Across the Calciner and Therefore Did Not Decrease the Flow of Gasses Through the Pyroprocessing System

CEMEX's Contentions

"[T]his modification would have also increased the static pressure and reduce [sic] the air flow through the calciner."

Hofmann, p. 5-2.

"Reducing the cross section and increasing turbulence would cause the static pressure drop across the calciner to increase. At a given ID fan power consumption (amp load), this would reduce the volume of air that the fan can pull through the system and accordingly reduce the total amount of fuel that could be consumed in the pyroprocess. This would cause an overall decrease in fuel NOx emissions as well as a decrease in thermal NOx emissions in the kiln."

Young, Summary of Opinions, Opinion No.1, p. 5.

"Reducing the cross section and increasing turbulence would cause the static pressure drop across the calciner to increase."

Young, p. 35.

"FCT was likely aware that their suggested efficiency improvement would increase the static pressure and reduce the air flow rate through the calciner."

Young, p. 36.

"[T]he total amount of fuel that could be burned in the overall pyroprocessing system would have been reduced because of the reduction in air flow due to the static pressure increase across the calciner."

Young, p. 36.

Response

Both Mr. Hofmann and Mr. Young assume that the static pressure drop across the calciner increased. Neither expert has identified documents⁴ or provided process measurements and/or calculations that would support their contention that a change, if any, in the static pressure drop across the calciner occurred as a result of the modification, or if it did occur, that such a change was of a magnitude that it would have the system-wide effect of reducing gas flow that each of them attribute to the modification.

In 1996, CEMEX hired Fuel and Combustion Technology International (FCT) to evaluate and redesign the internal surfaces of the calciner to reduce or eliminate the problem of incomplete combustion within the calciner. This incomplete combustion, known locally as “burnover,” was manifested by unacceptably high temperatures in the preheater cyclone and sometimes the conditioning tower. The FCT study report was provided to CEMEX in August 1996.⁵

As Mr. Young asserts, a reduction in the cross-sectional area of a calciner can increase the internal resistance to gas flow (as would be evidenced by an increase in the static pressure drop across the calciner).⁶ However, judicious designs can reduce the pressure drop to a minor consideration in operation of the unit. In fact, it appears to me from my review of the record that this is what occurred at Lyons. FCT studied two designs for changing the internal dimensions of the calciner’s combustion chamber to improve recirculating flow patterns. Modeling performed by FCT suggested that both designs seemed to achieve similar improvements in fuel/air mixing that would reduce or eliminate “burnover.” FCT concluded that the “parallel section” design would have a negligible impact on the pressure drop across the calciner.⁷ For this reason, the

⁴ Mr. Young’s only support for this statement is a reference to the deposition of Mr. John Lohr. Discussing the ID Fan, Mr. Lohr without explanation or elaboration asserts, “And, in fact, we did one thing that you’ve already discussed that made -- made the pressure drop greater, and that was what we did in the combustion chamber.” Deposition of John Lohr, October 11, 2010, TR at 71:2-5.

⁵ Fuel and Combustion Technology International, Southwestern Portland Cement Company, Physical Model Study of Precalciner, August 2, 1996, CMXLYO00453790 – 453839.

⁶ Young, Summary of Opinions, Opinion No.1, p. 5.

⁷ Case Study of Physical Modeling at Southdown’s Lyons Plant, CMXLYO00402635 – 402640.

“With this apparent success at hand[,] the FCT engineers decided to take the concept one step further and restrict the area above the burners even more. . . . This test did not indicate any substantial improvement over the previous parallel section insert, and with the added risk of restricting air flow *they decided against restricting the area beyond the parallel section.*”

“When FCT presented the final results of their modeling study to the Southdown corporate staff in the fall of 1996 they predicted a 7% improvement in production if the calciner was modified with their parallel section configuration. At the time there was some skepticism as well as some worry that the parallel insert would restrict air flow. FCT agreed to study the model further to predict the potential for increased pressure drop resulting from the insert. *When that study indicated negligible restriction we [CEMEX] started making plans to follow their recommendations.*”

Id. at CMXLYO00402637 (emphasis added).

“parallel section” design was selected and installed in the calciner by CEMEX in early 1997.⁸ Consequently, the gas flow rate through the calciner and kiln either would not be reduced or would only be impacted negligibly.

This assessment is consistent with the facility’s production history. CEMEX reported a 2.9 tons per hour (5.8%) increase in clinker production attributable to the calciner modification.⁹ In his deposition, Mr. John Lohr, the plant manager in 1997, confirmed the accuracy of this production rate increase.¹⁰ Elsewhere, even higher production rate increases have been attributed to the calciner modification.¹¹ In my opinion, it would be impossible to accomplish clinker production increases of this magnitude under conditions where gas flow through the calciner was reduced. By extension, if gas flow through the calciner was not reduced by this modification, there is no reason to believe the modification would reduce gas flow through the kiln system and its concurrent capacity to burn fuel.

B. The 1997 Modification of the Calciner Did Not Significantly or Permanently Alter the Distribution of Fuel Between the Calciner and Kiln

CEMEX’s Contentions

“Moreover, *to the extent* CEMEX was able to shift its fuel split in favor of the calciner after this improvement, *any such shift* would have further decreased thermal NOx formation in the kiln. Improved calcination in the calciner would also have lowered the thermal load on the kiln and further reduced thermal NOx.”

Young, Summary of Opinions, Opinion No.1, p. 5 (emphasis added).

“Approximately, 50-55% of total coal is combusted at calciner and remaining in the rotary kiln [sic], *although this percentage varied over time.*”

Hofmann, Summary of Opinions, Opinion No. 4, p. 1-3 (emphasis added).

“The rotary kiln’s thermal NOx *would have* lowered due to reduced air flow resulting from this project *if* the fuel split did not change and *would have* further reduced *if* the fuel split shifted to the precalciner.”

Hofmann, Summary of Opinions, Opinion No. 10, p. 1-4 to 1-5 (emphasis added).

“Therefore, *it is reasonable to assume* that after this project, fuel split between kiln and calciner *might have improved* (more in calciner vs. kiln). Mr. Goodrich is his testimony points out that the current fuel split between calciner and kiln is

⁸ Lohr Deposition Exhibit 203, Outline of Calciner Vessel with Parallel Section Above Burners.

⁹ Lyons Clinker Production History, May 17, 2000, CMXLYO00045592.

¹⁰ Deposition of John Lohr, October 11, 2010, TR at 40:5 – 41:1.

¹¹ PowerPoint, CEMEX, Inc., Lyons Cement Plant, CMXLYO00180055 at CMXLYO00180060 (“Kiln Throughput History, Combustion Chamber Modification, + 80 t/d (3.33 tons per hour or 6.7%)”); Lyons Plant – Monthly Report of Operations, April 1997, CMXLYO00049947 (“The calciner production chamber has resulted in a 10% increase in production on the kiln.”).

55-45%. Per Mr. Young's expert report, more fuel usage in calciner versus the kiln will reduce overall NOx emissions."

Hofmann, p. 5-3 (emphasis added).

Response

Both Mr. Hofmann and Mr. Young assume that the distribution of fuel between the calciner and kiln shifted permanently in favor of the calciner after CEMEX's 1997 modification to the internal geometry of the calciner's combustion chamber.¹² The conditional language emphasized in the above opinion excerpts highlights the uncertain nature of their contentions. It is in fact unsupported speculation. Neither expert has identified documents¹³ or provided measurements and/or calculations that would support their contention that a change, if any, in the fuel distribution occurred as a result of the modification, or if such a change did occur, was a consistent and permanent operational shift in the distribution of fuel between the calciner and kiln. Instead, as my opinion is presented below, it is likely that the fuel distribution at Lyons varied between 50:50 to 55:45 over time in response to operating conditions and that there was no clear and permanent shift in that distribution as a direct result of CEMEX's 1997 modification to the internal geometry of the calciner's combustion chamber. Mr. Hofmann acknowledges that the fuel distribution between the calciner and kiln varied over time¹⁴ although he does not reconcile that view with his and Mr. Young's contention that this was a permanent shift that contributed to a decrease in thermal NOx formation in the kiln.¹⁵

In an article on oxygen enrichment in 2001, CEMEX employees Herman Tseng and John Lohr report that "[t]he kiln is coal fired with approximately a 50:50 percent between main burner and calciner."¹⁶ The 50:50 fuel split is the most often identified relationship between calciner and kiln fuel distribution.¹⁷ As the fuel split is a variable that is controlled by the console operator based on operating conditions, historically at the Lyons Plant it has varied between

¹² I note that there seems to be some disagreement between Mr. Hofmann and Mr. Young as to the fuel split. See, for instance, Young Summary Opinion No 2, p. 6 (discussing fan rpms). "Also, any additional fuel that could have been used would have been split approximately 50% to the calciner."

¹³ Mr. Hofmann states that, "Mr. Goodrich is his testimony points out that the current fuel split between calciner and kiln is 55-45%." Hofmann, p. 5-3 n. 69. Mr. Goodrich's precise testimony is equivocal on this point. "Generally that the split, *as I understand from very limited discussions*, has fairly consistently been 55 percent calciner, 45 percent kiln range as it is today." Deposition of Steve Goodrich on December 9, 2010, TR at 42:9 – 12 (emphasis added).

¹⁴ Hofmann, Summary of Opinions, Opinion No. 4, p. 1-3.

¹⁵ "Per Mr. Young's expert report, more fuel usage in calciner versus the kiln will reduce overall NOx emissions." Hofmann, p. 5-3.

¹⁶ Herman Tseng and John W. Lohr, CEMEX, Inc., Oxygen Enrichment, International Cement Review, May 2001, CMXLYO00045538 at 45539; Herman Tseng & John W. Lohr, CEMEX, Inc. (formerly Southdown, Inc.), Oxygen Enrichment to Calciner Firing, CMXLYO00612164 at 612165 ("The kiln is coal fired with approximately a 50:50% fuel split between main burner and calciner.").

¹⁷ Herman Tseng and John W. Lohr, CEMEX, Inc., Oxygen Enrichment, International Cement Review, May 2001, CMXLYO00045538 at 45539. See also, Herman Tseng & John W. Lohr, CEMEX, Inc. (formerly Southdown, Inc.), Oxygen Enrichment to Calciner Firing, CMXLYO00612164 at 612165 ("The kiln is coal fired with approximately a 50:50% fuel split between main burner and calciner.").

50:50 and 55:45 (favoring the calciner).¹⁸ Thus, CEMEX's own documents demonstrate that for the period in question, 1997 – 2000, there was no clear and permanent shift in that distribution as a direct result of CEMEX's 1997 modification to the internal geometry of the calciner's combustion chamber.

A change in the percentage of fuel burned in the calciner could result in changes in the relative quantities of fuel and thermal NOx generated in the system. However, within the narrow range of the fuel shift under consideration, it is most likely that the total amount of fuel burned has more influence on the total amount of NOx generated than a fuel shift. Without a significant increase in thermal efficiency,¹⁹ an increase in clinker production generally will require that more fuel is burned, which in turn implies that more NOx will be generated.

C. No Evidence Exists to Support the Contention that the 1997 Modification to the Calciner Resulted in Increased Calcination in the Combustion Chamber

CEMEX's Contentions

"This project would have also improved the calcination process (more of calcination process being performed in the calciner vs the kiln)."

Hofmann, p. 5-3.

"Improved calcination in the calciner would also have lowered the thermal load on the kiln and further reduced thermal NOx."

Young, Summary of Opinions, Opinion No.1, p. 5.

"Improved combustion conditions in the calciner would have resulted in a higher degree of calcination, reducing the heat (fuel) requirements in the rotary kiln and thereby reducing thermal NOx formation."

Young, p. 36.

Response

The assertions made by Mr. Young and Mr. Hofmann regarding increased calcination resulting from the 1997 modification to the calciner's combustion chamber are contradicted by the record. The degree of calcination of the material leaving the Lyons calciner has historically been in the 70-75% range.²⁰ Oxygen enrichment and the reconfiguration of the combustion chamber did *not* change the amount of calcination that took place in the kiln. In their 2001

¹⁸ See, e.g., Daily Production Notes, October 29, 1999, CMXLYO00616602, ("Remember the fuel split is to be 41-48% on the kiln (43-50% LA), 52-59% on the tower (50-57% LA)") (emphasis omitted). Compare and contrast the foregoing with Lyons Plant Pyroprocess Training, CMXLYO00186916 – 187061 at 187032 ("Must balance primary and secondary fuel and O2. For precalciner: 50-60% precalciner fuel (heat basis); 40-50% main fuel (heat basis).").

¹⁹ See, Rebuttal Report of Dr. Phyllis Fox, June 24, 2011.

²⁰ Herman Tseng and John W. Lohr, CEMEX, Inc., Oxygen Enrichment, International Cement Review, May 2001, CMXLYO00045538 at 45539.

article, Mr. Tseng and Mr. Lohr confirm that "... the degree of precalcination ... *remained* at 70-75%."²¹ As the degree of calcination within the accepted narrow range changes, the temperature profile in the pyroprocess may vary slightly; however, there is more than enough thermal energy to accomplish the calcination either in the kiln using heat left over from clinkering or in the calciner using the leftover heat and added heat from burning fuel. While important to successful kiln operation, the range of the degree of calcination of the material leaving the calciner is a characteristic of the kiln system and a consequence of how the kiln system is operated. It is not a closely controlled operating variable. The degree of calcination is dependent on other operating variables such as the operating temperature of the calciner and the feed rate of raw mix to the calciner.

D. No Evidence Exists to Support the Contention that the 1997 Modification to the Calciner's Combustion Chamber Resulted in a Reduced Kiln Thermal Load

CEMEX's Contentions

"In March 1997, the calciner combustion chamber was improved to solve a historical 'burnover' problem in which calciner fuel was burned and effectively wasted downstream of the calciner. . . . As discussed in this report, overall fuel based NOx generation in the calciner most likely decreased as a result of the combustion improvement, or, to the extent it increased, did so by decreasing the amount of fuel combusted in the rotary kiln. In both scenarios, the effect of the change would have been a reduction in total NOx emissions from the pyroprocess. The rotary kiln's thermal NOx would have lowered due to reduced air flow resulting from this project if the fuel split did not change and would have further reduced if the fuel split shifted to the precalciner. Accordingly, this project would not have caused any NOx increases. Therefore, it was reasonable for CEMEX to assume at that time that no PSD review was required for this project."

Hofmann, Summary of Opinions, Opinion No.10, p. 1-4 to 1-5.

"During 1997, CEMEX made two efficiency improvements to their one-stage preheater, calciner pyroprocessing system. One improvement was made to the calciner internal geometry in order to reduce 'burnover.' The improvements to the calciner internal geometry were made in the spring of 1997 and reduced the cross sectional area of the calciner to increase turbulence for better air-fuel mixing. Reducing the cross section and increasing turbulence would cause the static pressure drop across the calciner to increase. At a given ID fan power consumption (amp load), this would reduce the volume of air that the fan can pull through the system and accordingly reduce the total amount of fuel that could be consumed in the pyroprocess. This would cause an overall decrease in fuel NOx emissions as well as a decrease in thermal NOx emissions in the kiln. Moreover, to the extent CEMEX was able to shift its fuel split in favor of the calciner after

²¹ *Ibid.* (emphasis added).

this improvement, any such shift would have further decreased thermal NOx formation in the kiln. Improved calcination in the calciner would also have lowered the thermal load on the kiln and further reduced thermal NOx.”

Young, Summary of Opinions, Opinion No.1, p. 5.

“It would be reasonable to conclude that this improvement to the calciner would decrease overall NOx emissions for several reasons . . . With less fuel being burned in the rotary kiln, thermal NOx would decrease.”

Young, p. 36.

“Improved combustion conditions in the calciner would have resulted in a higher degree of calcination, reducing the heat (fuel) requirements in the rotary kiln and thereby reducing thermal NOx formation.”

Young, p. 36.

Response

As I have explained in 2.A. – 2.C. above, there is no evidence that the 1997 modification to the calciner’s combustion chamber (1) significantly increased static pressure drop across the calciner, (2) permanently altered the calciner/kiln fuel split, or (3) increased the degree of feed calcination in the calciner. Therefore, to the extent that CEMEX’s expert opinions rely on those propositions to conclude that the 1997 modification to the calciner’s combustion chamber lowered the thermal load on the kiln and further reduced the generation of thermal NOx, their arguments fail. No evidence of such a decrease has been presented. Even had these relatively small changes in the process parameters occurred at a constant rate of production (which they did not), the changes discussed in 2.A. – 2.C. above would not have altered significantly the amount of thermal work that was required in the kiln to form acceptable clinker. The NOx emissions associated with that workload would not, then, have been reduced.

As Mr. Young concedes²² (despite his predicted effects of the alleged pressure drop across the calciner), the calciner modification allowed more feed to be processed through the pyroprocessing system, i.e., more clinker production. The pyroprocessing system’s thermal workload would have increased proportionately and would be expected to result in increased NOx emissions from one or both combustion locations. As has been noted numerous times, CEMEX anticipated and attributed increased clinker production to the combustion chamber modifications. All other things being equal, more clinker production requires more fuel combustion that in turn produces more NOx. Absent a very large improvement in overall thermal efficiency or the application of a NOx reduction technology, neither of which occurred at Lyons, additional clinker cannot be produced without potentially generating more NOx. The issue then becomes “how much” additional NOx was produced.

²² Young, p. 36 (“CEMEX could process more feed and produce more clinker with the same quantity of fuel.”). See also, Deposition of John Lohr, October 11, 2010, TR at 28:16 – 25; Lyons Clinker Production History, May 17, 2000, CMXLYO00045592.

Not previously mentioned in these discussions of potential effects of these projects is the fact that for some calciner kilns, if carbon monoxide (CO) emissions decrease, then NOx emissions increase. Perhaps this possibility should not have been overlooked at Lyons by CEMEX or its experts. One of the primary purposes of the changes to the Lyons calciner was to reduce the CO generation and emissions that resulted from the incomplete combustion associated with “burnover.”²³ In other words, if successful, CEMEX’s calciner modification that reduced CO emissions could have been counterproductive as far as NOx emissions were concerned. The reason is because the poor combustion conditions that produce CO and “burnover” also produce the free radicals that react with NOx and destroy it.

E. CEMEX Should Have Considered Whether the 1997 Modification to the Calciner’s Combustion Chamber Triggered PSD Requirements

CEMEX’s Contention

“The U.S. EPA’s NOx control document predicts a worst-case fuel NOx emission factor of 1.5 lbs of total fuel NOx/ton of clinker. Considering 50-50% split between kiln and precalciner fuel at the Lyons plant before implementation of this project, the fuel NOx factor will be equal to 0.75 lb of fuel NOx at calciner or kiln/ton of clinker. Even ignoring any fuel efficiency improvements, if one would have considered worst-case 10% potential clinker production increase from this project, fuel NOx increase from the calciner alone would have been well below 40 tons of PSD significance threshold (16.48 tons of NOx = 43,949 tons of clinker increase*0.75 lb of NOx/ton of clinker/2,000). Therefore, it is reasonable to assume that no PSD evaluations were needed at that time. Even if one considers that all fuel NOx (kiln and calciner) would have changed as part of this project, net increases using 1.5 lb per ton overall fuel NOx factor, would still have been well below 40 tons. It is also reasonable to assume that either thermal NOx in the kiln would have remained unchanged or decreased.”

Hofmann, p. 5-4 to 5-5.

Response

The U.S. EPA’s NOx control document Mr. Hofmann references was not published until 2000, well after CEMEX made the calciner improvement so this fuel NOx factor, even if valid, could not have entered into CEMEX’s 1997 PSD evaluation of the calciner modifications.²⁴

Contrary to Mr. Hofmann’s assertion, the referenced document does not purport to present a worst-case fuel NOx emissions factor. Instead, it presents an illustrative example and a

²³ See, CEMEX Inc., Lyons, Colorado, Responses to EPA’s June 6, 2007, CAA Section 114 Request, CMXLYO00393767 at 393774.

²⁴ Battye, Rebecca et al, EC/R Incorporated, NOx Control Technologies for the Cement Industry: Final Report, United States Environmental Protection Agency, September 19, 2000, http://www.epa.gov/ttnnaqs/ozone/ctg_act/200009_nox_epa457_r-00-002_cement_industry.pdf.

subsequent illustrative calculation of how a fuel NO_x emission factor might be determined. In fact, the example calculation referenced by Mr. Hofmann uses a 10% conversion of nitrogen in the fuel to NO_x. Also included in the same paragraph of the report is an illustration using a 100% NO_x conversion factor. The document does not indicate what a proper or typical NO_x emission factor might be. The concluding sentence of the referenced paragraph starts with “[t]hus, even with only 10 percent conversion of coal nitrogen to NO_x...,”²⁵ thereby suggesting that the example is a minimum (best case) rather than a maximum (worst case) conversion. In addition, Mr. Hofmann does not explain why he is ignoring thermal NO_x in his analysis of the “worst-case” fuel NO_x emission factor.

Moreover, this conversion factor is not intended to be for general use in the cement industry and it certainly is not specific to the Lyons plant in 1997. For that period of time, if the coal usage factor in pounds of coal per ton of clinker and the nitrogen content of the coal were known, and *speculation* was made about the conversion factor for the fuel-based nitrogen to NO_x, Battye’s work could be more or less duplicated – but it still would be *speculation*.

To evaluate the applicability and validity of a calculated fuel NO_x factor specific to the Lyons Plant answers to the four following questions would have to be known: (a) How consistent was the fuel mix (gas and coal)? (b) How consistent was the source of the coal supply? (c) How consistent was the coal usage factor? (d) How consistent was the coal analysis? And (e) How much fuel nitrogen and what percent of that is converted to NO_x? Mr. Hofmann does not provide that information. Without attempting to consider these site-specific factors, Mr. Hofmann’s calculations are meaningless.

Further, Mr. Hofmann does not explain why he believes it would be reasonable for anyone to assume that thermal NO_x would have remained unchanged or decreased if the clinker production was increased by 10%.

3. THE CALCINER COMBUSTION IS “FLAMELESS”

CEMEX’s Contentions

“[F]ossil-fuel-fired flames are required in the rotary kiln (*although not in the calciner*).

Davis, p. 3 (emphasis added).²⁶

“Prior to the use of OE, the coal burners in the calciner were supplying the heat for the highly energy demanding calcination reaction occurring in the raw mix and, as indicated by the problem with continued hydrocarbon/CO reactions in the preheater, were not resulting in complete combustion. The addition of oxygen into the tertiary air stream *would be expected to enhance the stability/attachment*

²⁵ *Id.* at p. 31.

²⁶ See, Project Outline for Oxygen Enrichment Test at Lyons, March 1997, CMXLYO00402688, which states that combustion in the calciner is flameless.

of the coal flames and, at these relatively mild temperatures, where thermal NOx formation rates would be very low, it would be reasonable to assume that, for a given coal firing rate, both overall NOx and hydrocarbon/CO emissions could be reduced while clinker production increased.”

Davis, p. 13 (emphasis added).

[Referring to the calciner] “The terminology ‘burner’ in this instance could be confusing as most burners also have closely associated ‘combustion air.’ *The term fuel injector would be a reasonable term as well.*”

Davis, p. 13 n. 45 (emphasis added).

“The modifications made to the refractory immediately above the burners should have resulted in improved recirculation of hot coal combustion products to the root of the flame for improved flame attachment and stability. As with OE, the improved flame stability should result in less intimate mixing of the oxygen-containing gases into the base of the fuel-rich NOx reducing region. Again, it is reasonable to expect this modification to result in improved fuel efficiency and clinker production with lower emissions of hydrocarbons, CO and NOx.”

Davis, p. 15 (emphasis added).

“In a calciner, no defined flame exists. Fuel is oxidized under ‘flameless’ conditions.”

Young, p. 26 (emphasis added).

Response

The Project Outline for Oxygen Enrichment Test at Lyons - March 1997, describes combustion in the calciner as follows: “*low-temperature flameless combustion* in the precalciner.”²⁷ That description coincides with Mr. Young’s understanding that “*in a calciner, no defined flame exists.*”²⁸ Initially, Dr. Davis *appears to agree* with Mr. Young *as to the nature of combustion within the calciner, i.e.,* that “. . . fossil-fuel-fired flames are required in the rotary kiln (*although not in the calciner*).”²⁹ However, later Dr. Davis appears to contradict both himself and Mr. Young, when, in describing the calciner, he says, “The modifications made to the refractory immediately above the burners should have resulted in improved recirculation of hot coal combustion products *to the root of the flame for improved flame attachment and stability.*”³⁰ In light of this contradiction between CEMEX’s own experts, and given that Dr. Davis does not provide references to the record or provide process measurements and/or calculations to support

²⁷ Project Outline for Oxygen Enrichment Test at Lyons, March 1997, CMXLYO00402688.

²⁸ Young, p. 26 (emphasis added).

²⁹ Davis, p. 3 (emphasis added).

³⁰ Davis, p. 15 (emphasis added).

his observations, I contend that his observations on flame attachment, and consequently his conclusions on NOx emission reductions, are unsubstantiated.

4. REPLACEMENT OF THE ID FAN MOTOR AND THE RELATED INCREASED ROTATIONAL FAN SPEED SHOULD HAVE CAUSED CEMEX TO CONSIDER WHETHER PSD REQUIREMENTS WERE TRIGGERED

CEMEX's Contentions

"The plant's 1,250 hp ID fan motor failed in August 1997 resulting in an emergency shutdown of the kiln. CEMEX management replaced the failed motor with the only readily available motor with a RPM rating matching the existing ID fan. The ID fan and associated fan curve were not changed. At the time of this motor's replacement, it might have met the PSD's Routine Maintenance, Repair, & Replacement (RMRR) exclusion. Due to slightly increased motor HP and associated fan's RPMs, there was a possibility of a disproportionately lower increase in kiln flow and clinker production. Due to vibration and system resistance problems, it is reasonable to assume that CEMEX might not have realized any of this minimal increase[,] which is the conclusion reached in the expert report of Mr. Gerald Young. Regardless, considering a maximum 28 tons per day potential clinker increase and a baseline (August 1995 – July 1997) NOx emission factor of 7.83 lbs/ton, NOx emission increases would have been below 40 tons. Therefore, it was reasonable for CEMEX to have assumed at the time that PSD review was not triggered."

Hofmann, Summary of Opinions, Opinion No. 12, p. 1-5.

"Considering the situation surrounding this replacement and due to the fact that the fan and fan curve were not changed, at the time this motor was replaced, it would have been reasonable for CEMEX to assume that no extra RPMs/will be gained from the replacement."

Hofmann, p. 5-7.

"In 1997, the 1250 hp ID fan motor suffered a catastrophic failure. The pyroprocessing system was shut down because the ID fan is an integral part of the system. This constituted an emergency repair, not a planned project to increase clinker production. Any increase in rpm and air flow that was realized at the location of the fan because of the larger motor would have resulted in a significantly smaller increase in air flow through the pyroprocessing system due to the fact that the air flow rate at the fan is not fully realized at the fuel firing locations. For example, ambient air leaks and water sprays in the conditioning tower will make up a part of any increase in air flow through the fan. Also, any additional fuel that could have been used would have been split approximately 50% to the calciner. The effect on total NOx emissions would have been minimal. Based on the author's experience, plant personnel would have been

reasonable not to have considered the emergency replacement of the ID fan motor to be a project to increase production or a PSD triggering modification.”

Young, Summary of Opinions, Opinion No. 2, p. 6.

“A 3.88% increase in the gas flow rate through the fan does not mean that 3.88% additional fuel could have been burned with a corresponding 3.88% increase in clinker production [T]he net available increase in the air flow rate that could support additional production is most likely in the range of 1.5 to 2.5%, rather than the 3.88% of gases passing through the ID fan.

Young, p. 37 – 38.

“If the kiln historical production rate of 54.5 tph (as of 1/4/1998) is used then the clinker production rate would be 1308 tons per day and 28 tons per day would be a 2.14% increase. This small increase would not, in the author's experience, have alerted the Lyons staff that it might significantly increase NOx emissions.”

Young, p. 38.

Response

A. The Replacement of the ID Fan Motor and Operation of the ID Fan at Increased Rotational Speeds Contributed to Significant Increases in NOx Emissions

At pages 28 – 29 of my expert report in this matter, I concluded that the replacement of the induced draft (ID) fan motor at the Lyons plant in late August 1997 and CEMEX’s decision to run the fan at a higher speed increased the maximum potential volumetric flow rate of gas through the pyroprocessing system. I also noted that CEMEX ran the new ID fan motor at higher rotational speed and noted that CEMEX attributed a clinker production increase of an additional 12,902 tons per year to the 1997 replacement of the ID fan motor. I further concluded that the increase in gas flow attributable to the new fan motor allowed additional fuel combustion in the kiln and calciner and as a result, there would have been a significant increase in NOx emissions from the pyroprocessing system.³¹ Both Mr. Hofmann and Mr. Young disagree with these conclusions. As explained below, their positions are incorrect.

In evaluating the suitability of the proposed 1,500 horsepower motor before the purchase was made, a responsible person within CEMEX would have compared the new motor characteristics to the specification sheet and the fan curve for the existing fan to make certain that the two devices were compatible. A very critical element to check would have been the 1,150 revolutions per minute maximum design speed of the fan.³² It is at this speed that the fan will not move any additional gas no matter how much faster it turns. If the fan performed

³¹ Expert Report of Walter L. Greer, United States v. CEMEX, Inc., Civil Action No. 09-cv-00019-MSK-MEH in the U.S. District Court for the District of Colorado, March 1, 2011, p. 28 – 29 (hereinafter “Greer”).

³² See, Zurn Industries, Inc., Air Systems Division, Fan Specifications, May 8, 1979, CMXLYO00628865.

satisfactorily with a 1,250 horsepower motor, a 1,500 horsepower motor also would have been satisfactory if the motor's rotational speed was appropriate, which clearly it was. With the old motor, the ID fan ran at a maximum rotational speed of about 1,030 revolutions per minute.³³ Applying the third fan law³⁴ in which rotational speed is related to horsepower as shown below, it could have been calculated that the increased available horsepower would support a higher speed of about 1,094 revolutions per minute, a speed well below the maximum design speed for the fan. Therefore, the new motor would support a higher fan speed.

$$\begin{aligned}\text{New RPM} &= (\text{New HP/Old HP})^{0.33} \times \text{Old RPM} \\ \text{New RPM} &= (1500/1250)^{0.33} \times 1030 = 1094\end{aligned}$$

The first fan law³⁵ says that an increase in volume through the fan is directly proportional to the increase in rotational speed; therefore, an increase in flow volume should have been anticipated through the use of the new motor and at an increased rotational speed of the fan.

$$\begin{aligned}\text{New CFM/Old CFM} &= \text{New RPM/Old RPM} \\ \text{New CFM} &= (\text{New RPM/Old RPM}) \text{ Old CFM}\end{aligned}$$

Shortly after resuming kiln operation with the new motor in place, CEMEX increased the speed of the induced draft fan to about 1,070 revolutions per minute,³⁶ a value less than that projected by the fan law because of vibration issues with the fan.³⁷ Nevertheless, CEMEX doubtless increased the fan speed to its practical upper limit knowing that the volume of gas handled by the fan would increase to support an unknown but increased, bonus rate of clinker production. Otherwise, the increased fan speed would have been pointless. Alternatively, CEMEX could have held in reserve the potential increase in fan speed to provide the kiln operator another mechanism to better cope with upsets in the pyroprocess.

In considering the increase in clinker production achieved through the increased gas flow in the kiln system, I did not attempt to identify or quantify the origin of the increased gas flow, *i.e.*, combustion products, calcination products, water vapor, or air leaking into the system, and thereby calculate a potential increase in clinker production. That would be a rather difficult process engineering exercise as demonstrated by Mr. Young's lack of specificity in his hypothetical example in Section 4.1.2 of his report.³⁸ Rather, I took the word of plant manager Mr. Lohr, who has affirmed the accuracy of his estimate of increased clinker production as a result of the new induced draft fan motor.³⁹ As the most accurate contemporary estimate, I used Mr. Lohr's increased clinker production rate in the screening test that I performed in my expert report only to identify the need for a more detailed analysis of regulatory applicability of the fan motor replacement. Using that approach, I concluded that the potential increase in NOx

³³ See, Deposition of Steve Goodrich, December 9, 2010, TR at 73:14 – 20.

³⁴ See, Young, Section 2.15, p. 21 – 22.

³⁵ *Ibid.*

³⁶ See, Deposition of Steve Goodrich, December 9, 2010, TR at 95:22 – 96:1; *see also*, Rebuttal Report of Dr. Phyllis Fox, June 24, 2011.

³⁷ See, Deposition of Steve Goodrich, December 9, 2010, TR at 78:12 – 79:1.

³⁸ See, Young, May 10, 2011, Section 4.1.2, p. 37 – 38.

³⁹ See, Lyons Clinker Production History, May 17, 2000, CMXLYO00045592; Deposition of John Lohr, October 11, 2011, TR at 31:11 – 13.

emissions from the ID fan motor replacement and the increased rotational speed of its operation would have been approximately 56.1 tons per year.

There appears to be universal agreement among the experts that there was an increase in potential NOx emissions resulting from the replacement of the ID fan motor and the subsequent increase in speed of the ID fan. Prior to or shortly after the speed of the replacement fan was increased, CEMEX should have considered if there was a “red flag” for NOx emissions resulting from this event that would require further consideration for PSD applicability. It was only important that a need for a permit was not overlooked - not that an absolute value was assigned to potential NOx emissions. The approach that I used in my expert report was conservative but safe and indicated that PSD implications should have been further considered, particularly in light of the contemporary projects that were intended to increase clinker production. My value for a potential NOx emissions increase was 56.1 tons per year.⁴⁰

There are many ways to calculate this potential NOx emission increase. Mr. Young simply used a smaller clinker production rate than Mr. Lohr had assigned to the event (which I used in my calculation) and only subjectively concluded that the 2.14% clinker production rate increase was too small to have generated potentially 40 tons per year of NOx emissions.⁴¹ Mr. Young did not consider an annual NOx emission multiplier in reaching his conclusion. If Mr. Young only had used the actual 1996 NOx emission rate of 1,873 tons⁴² for the previous year and his 2.14% clinker production increase, it can be shown that the potential NOx increase could have been calculated to 40.08 tons per year, a regulation-significant amount.⁴³ If Mr. Young had used Mr. Lohr’s slightly higher 3.0% increase in clinker production and the 1996 NOx annual emission rate of 1,873 tons, then the potential increase in NOx emissions would have been 56.2 tons per year.

In searching for a lower clinker production multiplier, Mr. Hofmann dismisses Mr. Lohr’s 3.0 clinker production increase because it was compiled in 2000, “many years after this project.”⁴⁴ In his deposition, however, Mr. Lohr affirmed the value of the clinker production increase that he assigned to the replacement of the ID fan motor.⁴⁵ Interestingly, in an apparent search for a smaller clinker production multiplier of 9,408 tons per year, Mr. Hofmann uses a later assessment of the clinker production increase—a number that was compiled in “2001 or 2002” and uses a lower NOx emission factor of 7.83 pounds of NOx per ton of clinker to conclude that “this project would have resulted in an emission increase of at most 36 tons....”⁴⁶ Actually there is a rounding error as his calculated increase should have been 37 tons of annual NOx.⁴⁷ Should Mr. Hofmann have chosen to use Mr. Lohr’s more contemporaneous figure 12,902 tons per year estimate of the increase in clinker production attributable to the larger ID

⁴⁰ Greer, p. 29.

⁴¹ Young, p. 38.

⁴² NOx vs Clinker Production (1996 – 2008), CMXLYO00149592 at 149594.

⁴³ $(1,873) \times (0.0214) = 40.08$

⁴⁴ Hofmann, p. 5-8.

⁴⁵ Deposition of John Lohr, October 11, 2010, TR at 31:11 –13.

⁴⁶ Hofmann, p. 5-9.

⁴⁷ $[(9,408) \times (7.83)] / 2000 = 36.8$

fan motor and still used his NOx emission factor of 7.83 pounds of NOx per ton of clinker, the annual increase in NOx emissions would have amounted to 50.5 tons.⁴⁸

Regardless of the calculation method chosen, CEMEX should have been aware that the increased clinker production made possible by the new fan motor would result in a substantial increase in NOx emissions and assessed PSD applicability with some rigor. Based on my review of the record, no such assessment was conducted.

B. Neither Replacement of the ID Fan Motor nor Operation of that Motor at Increased Rotational Speed Can be Considered Routine Maintenance, Repair and Replacement

CEMEX's Contentions

"At the time of this [ID Fan] motor's replacement, it might have met the PSD's Routine Maintenance, Repair, & Replacement (RMRR) exclusion."

Hofmann, Summary of Opinions, Opinion No. 12, p. 1-5.

"As such, the U.S. EPA made the case that applicability of the RMRR exclusion was to be evaluated by the source on a case-by-case basis by weighing the nature, extent, purpose, frequency, and cost of the work, as well as other relevant factors to arrive at a common sense finding."

Hofmann, p. 5-9.

Response

No doubt that the failure of the Lyons ID Fan in late 1997 was a significant event. However, when I apply the factors identified by Mr. Hofmann to the record at hand, I come to a different conclusion regarding the RMRR exclusion that he discussed. The "nature" of the replacement was that in 1997, the existing 1,250 HP motor for the kiln ID fan failed and was replaced by an available 1,500 HP motor.⁴⁹ In that regard, clearly the ID Fan replacement was a physical change. Moreover, the added capacity resulting from the increase in motor size from 1,250 HP to 1,500 HP was significant - the design specification listed a fan speed of 1,150 RPM.⁵⁰

As the plant manager testified, the ID Fan was badly damaged:

⁴⁸ $[(12,902) \times (7.83)] / 2000 = 50.5$. See also, Rebuttal Report of Dr. Phyllis Fox, June 24, 2011.

⁴⁹ See, Southdown, Inc., Kiln Volatile Management, Lyons Colorado, March 20, 2000, CMXLYO00045611 at 45620; Letter from John Lohr to Emergency Breakdown Team, August 11, 1997, CMXLYO00453897.

⁵⁰ Zurn Industries, Inc., Air Systems Division, Fan Specifications, May 8, 1979, CMXLYO00628865.

“[The ID fan] was pretty badly damaged, as I recall, and so their estimate was either take it all the way down to the frame, total rewind, total rebuild on the motor.”⁵¹

While the ID fan motor could have been repaired within 6 – 7 days, because the timeframe for repair was uncertain, CEMEX replaced the fan in its entirety.⁵² The “extent” of the action therefore was a total replacement of the motor.

The “purpose” of the fan motor replacement is clear. Without an ID fan the kiln could not be operated. There also is no confusion in the record that CEMEX quickly decided to take advantage of the additional horsepower of the replacement motor to increase the ID fan speed to its practical limit. No matter the intent or circumstances of the ID fan motor replacement, the effect of the larger motor was to increase the potential clinker production capacity of the Lyons pyroprocessing system.

The 1,250 HP fan motor was installed when the kiln was modified, sometime between 1979 – 1980.⁵³ Given that the motor had a 10 – 20 year operating life,⁵⁴ at the time of the failure it was at or beyond its expected service life. As the present plant manager, Mr. Goodrich, testified, replacement of these motors is “infrequent”⁵⁵ but replacements will occur over the long life of a cement plant.

Finally, while the “cost” of the replacement motor is relatively insignificant, failure to replace or repair the motor ultimately would have meant shutting down kiln operations entirely for a week, which is a costly proposition for any plant. It will be for others to apply the WEPCO factors Mr. Hofmann discusses, but from my experience, this type of a situation is not routine. As the plant manager said in a thank you note to the repair team, “This event created an emergency situation at the plant.”⁵⁶

There is no dispute that the failure of the induced draft fan motor was an emergency situation (although it could have been anticipated and a spare motor identified in advance) and that CEMEX selected the most economically attractive option available at that time to replace the failed motor. However, after the immediate problem was resolved and when the subsequent decision was made by plant management to run the replacement motor and the fan at the maximum possible speed, whether limited by motor current or fan vibration, the regulatory implications should have been considered. At least two possible alternatives were available to the decision makers within CEMEX: 1) limit the speed of the new fan to the old value and obtain

⁵¹ Deposition of John Lohr, October 11, 2010, 30:10 – 13.

⁵² *Id.*, TR at 65:25 – 66:17. *See also*, Letter from John Lohr to Emergency Breakdown Team, August 11, 1997, CMXLYO00453897.

⁵³ Deposition of John Lohr, October 11, 2010, TR at 62: 13 – 22. *See also*, Colorado Department of Public Health and Environment – Air Pollution Control Division (CDPHE – APCD), Summary of Preliminary Analysis for Best Available Retrofit Technology, September 1, 2008, CMXLYO00133622 at 133624; CDPHE – APCD, Preliminary Analysis, Permit No. 12BO444-2, March 6, 1989, CMXLYO004525024 at 425042 – 425043.

⁵⁴ Deposition of John Lohr, October 11, 2010, TR at 62: 21 – 63:1. Deposition of Steven Goodrich, December 9, 2010, TR at 76:25 – 77:10.

⁵⁵ Deposition of Steven Goodrich, December 9, 2010, TR at 77:11 – 13.

⁵⁶ Letter from John Lohr to Emergency Breakdown Team, August 11, 1997, CMXLYO00453897.

a federally enforceable limitation from CDPHE, or 2) perform an analysis of the effect of the higher fan speed on clinker production and on the potential emissions of the plant, and, if indicated, seek an applicability determination from the CDPHE prior to raising the speed. CEMEX chose neither of these options and simply unilaterally increased the speed of the fan and the potential clinker capacity of the Lyons kiln system.

5. MARCH 1997 OXYGEN ENRICHMENT TEST

CEMEX's Contention

"The only way CEMEX management could have determined the impact of OE on NO_x overall kiln system was to assess the actual NO_x emission rate while oxygen was injected and when it was not."

Hofmann, p. 5-11.

Response

The proper way to execute a test to predict the effect of OE would have been to conduct a test under those conditions at which the pyroprocessing system was anticipated to run in the future. In addition, if NO_x emissions were to be assessed, the operating parameters that contribute to NO_x formation and emission should have been held as constant as possible. Otherwise, CEMEX would have developed false and misleading information on the impact of OE that was not predictive of the future behavior of NO_x emissions from the Lyons kiln. One of the most critical and instructive of the relevant operating parameters is the electrical current drawn by the drive motor for the rotary kiln, *i.e.*, "kiln amps."

In CEMEX's pyroprocessing training, operators are instructed in the truth of the following parametric relationships:

"Bottom line: the **hotter the burning zone, the higher the NO_x**."⁵⁷

"Kiln Torque: **The hotter the [burning zone], the higher the amps**...the colder the [burning zone], the lower the amps"⁵⁸

CEMEX's pyroprocessing training manual elaborates further that a higher value for "kiln amps" indicates higher NO_x generation because the burning zone temperature is higher. The display of "kiln amps" is a useful and sensitive tool for operators because it is indicative of current conditions within and about the kiln. In addition to burning zone temperature and the resulting NO_x generation, a "kiln amps" measurement provides information on the condition of the coating within the burning zone and the mechanical condition of the kiln.⁵⁹ Nevertheless,

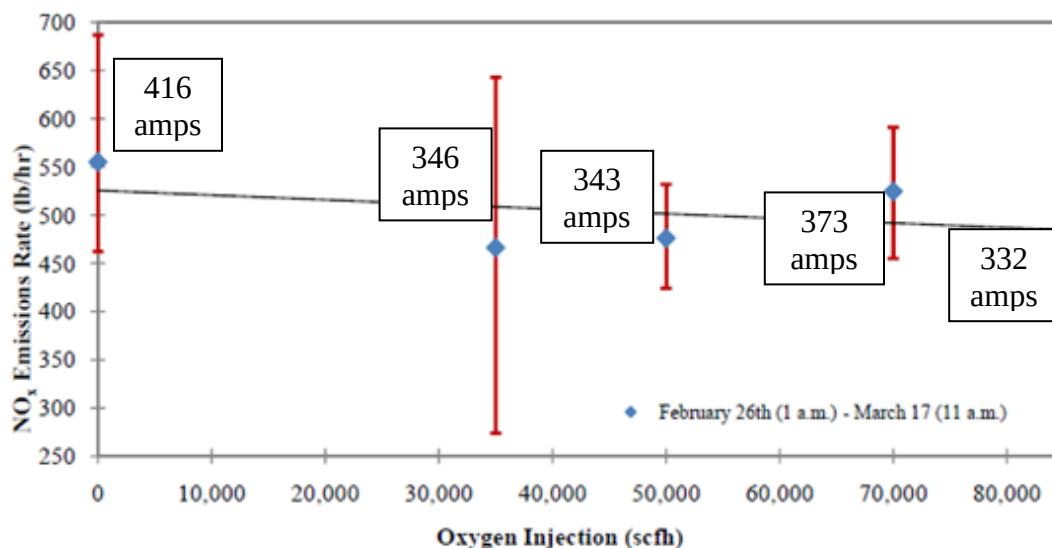
⁵⁷ Lyons Plant Pyroprocessing Training, CMXLYO00186916 at CMXLYO00186989 (emphasis added).

⁵⁸ *Id.* at CMXLYO00187038 (emphasis added).

⁵⁹ A portion of the melted material in the clinker "load" freezes onto the bricks that line and protect the steel shell of the kiln. This coating serves to protect the bricks from abrasion and thermal stress. If the coating is uneven, the kiln amps will vary during each revolution of the kiln because more or less energy is required to lift the kiln in different

observation of the changes in “kiln amps” is very useful in determining the direction of changes to the temperature in the burning zone of the kiln. As I will describe shortly, CEMEX appears to roughly equate a value of “kiln amps” to temperature within the burning zone. This practice is not uncommon in the industry.

FIGURE 5-1. 1997 OE TEST –PARTS 1 AND 2



On pages 5-11 to 5-12 of his expert report, Mr. Hofmann expresses the opinion that NO_x decreases as the oxygen injection rate increases. He attempts to demonstrate this relationship in Figure 5-1 reproduced above. However, when Mr. Hofmann’s Figure 5-1 is overlaid with kiln amps averages from each test period, a different picture emerges. It becomes clear that changes in “kiln amps” have a more direct relationship to NO_x emissions than the amount of oxygen being injected during the test. Contrary to Mr. Hofmann’s assessment, the test seems to have validated what was already well known, *i.e.*, the burning zone temperature directly affects NO_x emissions, rather than to have generated any new information about the effects of OE on NO_x emissions from the Lyons kiln system. In my opinion, the best that can be said is that OE had no effect on NO_x emissions from the Lyons pyroprocessing system during the 1997 test but even that observation is inconclusive primarily because of the inappropriately varying temperature in the hot zone of the rotary kiln. In evaluating the test results, it must be accepted that the most important aspect of the 1997 test was to determine the feasibility of OE with regard to economical clinker production and studying the effect of OE on NO_x emissions was no better than a secondary objective.

portions of each revolution. Mechanical problems such as a failing bearing also affect the amount of electrical power required to rotate the kiln.

6. APRIL 1998 OXYGEN ENRICHMENT TEST

CEMEX Contentions

“The duration of the 1998 test was sufficient for CEMEX to supplement previously tested NOx reductions due to OE injection and provided CEMEX with additional support for its belief that oxygen injection would not increase NOx emissions or trigger PSD.”

Hofmann, Summary of Opinions, Opinion No. 16, p. 1-6.

“It is my opinion that the 1998 test along with additional supporting information available at the time provided CEMEX with support that oxygen enrichment would not result in a significant emissions increase.”

Hofmann, p. 5-20.

“CEMEX also began using oxygen enrichment (OE) in the calciner to improve combustion conditions and to increase clinker production. CEMEX conducted tests in 1997 and 1998 to determine the effects of OE on emissions. Both tests indicated that NOx actually decreased. Accordingly, CEMEX was reasonable in concluding that it was not required to do a PSD review.”

Young, Summary of Opinions, Opinion No. 1, p. 5.

“It appears that CEMEX heated up the kiln system for this test series by instructing the console operators to target the kiln drive amps higher than normal, which is a reasonable way to stabilize the system in order to isolate the effects of the variables being tested.”

Young, p. 41.

Response

In my expert report on this matter at pages 42 – 45, I set forth my concerns with the reliability of the April 1998 oxygen enrichment test. It continues to be my opinion that the “Boardman” test was so poorly conceived, planned, executed, and reported that analysis of the results is an exercise in futility. Mr. Hofmann and Mr. Young dispute my reasoning. As discussed below, I disagree with their observations in this matter.

It is not unusual to operate a cement kiln system toward the “hot” side of its normal operating range on test day to ensure stability. A “hot” kiln is easier to operate than a “cold” kiln and the chances of catastrophic failure of the process (e.g., the exothermic reaction involved in clinker formation ceases) are much less with a “hot” kiln; therefore, the operator's preference on test day often is for a “hot” kiln even though such operation may not fully represent normal plant operating conditions.

Two operating parameters were measured and recorded on the Lyons kiln during the April 1998 test that were indicative of the temperature of the rotary kiln. These were “*Kiln Dr Amps*” (“kiln amps”) and “*Kiln Feed Hd Temp*” (gas temperature at the feed end of the kiln). All things being equal, these parameters are directly related to kiln burning zone temperature, i.e., when the kiln temperature increases, the values of these parameters also increase. Because of potential changes in the temperature profile of the kiln, short-term changes in the feed end gas temperature are not quite as indicative of burning zone temperature as kiln motor amps but this parameter has relevance nonetheless.

From the tables below and from Operating Reports (*see generally*, CMXLYO00085743-95758) the following facts can be observed. In the two days prior to commencement of the test, the kiln drive motor current in amps averaged 574.20 and 594.90, respectively. In the twelve hours *prior to the commencement of the “oxygen on” test*, the kiln amps averaged 469.3, thereby suggesting a decided cooling of the kiln prior to the test rather than the anticipated heating of the kiln as reported by Mr. Young. During the “oxygen on” portion of the test, the kiln amps averaged 437.28, suggesting a further cooling of the kiln with potentially lower NOx generation in the rotary kiln. However, during the “oxygen off” test that immediately followed the “oxygen on” phase, the kiln amps averaged 498.60, suggesting a heating of the kiln and potentially higher NOx generation in the rotary kiln during this portion of the test.

<u>Kiln Amps</u>	<u>12-Hrs Before Test</u>	<u>Oxygen On</u>	<u>Oxygen Off</u>
Hi	489.00	459.60	544.40
Lo	402.00*	381.60	469.20
Avg	469.30	457.28	498.60

If the operating data have been accurately recorded by CEMEX, the temperature differentials in the two test parts suggest a potential skewing of the test results to favor the conclusion that OE has the potential effect of lowering NOx emissions when actually it was the well known phenomenon of a direct relationship between burning zone temperature and thermal NOx emissions that was being observed. As previously explained in this rebuttal report, “kiln amps” is one of the operating parameters that kiln operators observe to evaluate the burning zone temperature. Specifically, prior to this test, CEMEX instructed its operators to cause the kiln amps to increase to a value of 500 when oxygen was off (a result of increasing burning zone temperature and thereby raising NOx) and to lower kiln amps back to a value of 400 where it had been when oxygen was on (a result of cooling the burning zone and thereby lowering NOx).⁶⁰ It is curious why CEMEX would issue these instructions for the test regarding “kiln amps” when it knew the relationship between “kiln amps,” burning zone temperature, and the generation of thermal NOx. All the variables, including “kiln amps” should have been held as constant as possible to properly evaluate the effect of OE.

From the tables below and from Operating Reports (*see generally*, CMXLYO00085743-95758), the following facts can be observed. In the two days prior to the test, the feed end

⁶⁰ Daily Production Notes, April 9, 1998, CMXLYO00085128 at 85129 (“When the test is over put additional dust on the kiln (6-8%), lower the CC temp, lower the feed hood temp, lower the drive amps to around 400, put the oxygen back on at 42,000, raise the ID fan back up if you lowered it.”).

temperature in degrees Fahrenheit averaged 1905.08 and 1853.17, respectively, and in the twelve hours prior to the commencement of the “oxygen on” test, the feed end temperature averaged 1858.00, thereby reinforcing my observation based on kiln amps that a cooling of the kiln occurred prior to the test rather than the heating of the kiln reported by Mr. Young. In the “oxygen on” portion of the test, the feed end temperature averaged 1838.40, suggesting a further cooling of the kiln and a potentially lower NOx generation in the rotary kiln during this portion of the test. During the “oxygen off” test that followed immediately, the feed end temperature averaged 1875.6, suggesting a heating of the kiln and potentially higher NOx generation in the rotary kiln. If the operating data have been accurately recorded by CEMEX, the feed end gas temperature differentials in the two tests substantiate a potential skewing of the test results to favor the conclusion that OE has the potential effect of lowering NOx.

<u>Feed Hood Temp</u>	<u>12-Hrs Before Test</u>	<u>Oxygen On</u>	<u>Oxygen Off</u>
Hi	1884	1882	1894
Lo	1816*	1790	1840
Avg	1858	1838**	1876

*Low value was in the one-hour period immediately before the "oxygen on" test commenced

** Average value for "oxygen on" lower than any value in "oxygen off" series.

All three of CEMEX’s expert rebuttal reports accept the commonly held knowledge that short term NOx emissions are highly variable, yet they inexplicably advance the notion that the April 1998 tests using only ten, two-hour increments split into two parts are sufficient in length to produce viable NOx emission results. In a paper written by Mr. Young,⁶¹ he states that a long-dry kiln on which he conducted research required 30.7 days of emission data to be statistically accurate to within 10% of the true emission rate. Similarly, he notes that fourteen days of emissions data were required to achieve the same statistical accuracy on a preheater kiln.⁶² The difference between twenty hours and one month is compellingly significant and suggests that a twenty-hour test on the Lyons kiln is inadequate even if important operating parameters could be held constant - which they weren't.⁶³ In the same paper, Mr. Young refers to an included graph representing 24 hours of NOx emissions data from a cement kiln in which the hourly average of six-minute data points for the NOx emission factor ranges from 2.24 to 11.72 pounds of NOx per ton of clinker. From this chart alone, it is intuitively obvious that short-term tests for NOx emissions from the Lyons kiln are problematic.

In Section 3.1 of the above referenced paper, *Operating Variability of Kilns*, Mr. Young describes an extreme circumstance in which a relatively small failure to increase the fuel firing rate on a timely basis could cause a kiln to suffer catastrophic process failure with attendant cyclical efforts to restore proper operation of the kiln.⁶⁴ He postulates that there would be rather large variations in NOx emissions resulting from this process failure. A similar but much less

⁶¹ Gerald L. Young, NOx Formation in Rotary Kilns Producing Cement Clinker, Applicable NOx Control Techniques and Cost Effectiveness of These Control Techniques, Section 3.2, “NOx Emissions Variability of Kilns,” p. 243, Institutes of Electrical and Electronic Engineers, 2002.

⁶² *Ibid.*

⁶³ Greer, p. 45.

⁶⁴ Young, NOx Formation in Rotary Kilns Producing Cement Clinker, Applicable NOx Control Techniques and Cost Effectiveness of These Control Techniques, Section 3.1, p. 242-243.

severe change in the Lyons kiln during the conduct of the April 1998 oxygen enrichment test can be seen on the Operating Report for April 9, 1998. Average kiln drive motor amps fell from 481.20 during hour 10 to a low of 381.60 in hour 14 but recovered to 482.40 in hour 17. These values suggest a kiln that cooled somewhat precipitously (*i.e.*, went “in the hole”) and then was restored to previous operating conditions. It is known that Type I raw mix also replaced Type II raw mix around hour 16 on this day. An abrupt change in feed chemistry, such as sometimes occurs when the type of clinker is changed, can cause an upset in kiln operation of the type and magnitude suggested by the operating data. During this part of the test period, other operating parameters show variability that may have had some effect on NOx emissions. In my opinion, the Boardman test definitely should have been aborted when the type of clinker being processed was changed and/or when the process upset occurred. About 30% of the brief test period was compromised by the upset. Over a long test period, *e.g.*, 30 days, the process variations and their effects on NOx emissions tend to average out. This smoothing of NOx data from a cement kiln is impossible to achieve in a twenty-hour test.

In the 1970s, after NOx CEMs were perfected and before NOx became a regulated pollutant, the NOx concentration in the flue gas of a cement kiln often was used as a process control tool because this operating parameter quickly indicated the relative temperature of the high-temperature, burning zone of the rotary kiln much faster than kiln amps or other measurements. Thermal NOx emissions are directly related to this temperature. When the kiln was getting hotter, the NOx concentration in the flue gas would increase. When the NOx concentration in the flue gas was falling, the kiln was getting colder. During the Boardman test, the other temperature indicator parameters were showing that the temperature and the resulting generation of NOx in the rotary kiln were changing independently of the variable allegedly being evaluated, OE. Therefore, in addition to its many other shortcomings, the Boardman test provided no relevant information on the effect of OE, if any, on the operation of the Lyons kiln system.

7. RISER DUCT FIRING

CEMEX’s Contentions

“A form of staged combustion, riser duct firing using coal, is employed at the Lyons Plant to improve operation of the calciner.”

Hofmann, p. 2-1.

“We are not aware of any documented explanation of precisely how oxygen injected into the calciner reduces the overall NOx generated in the process. A number of possibilities exist including, but certainly not limited to, the following...

A change in the fuel rich/fuel lean zones in the calciner[,] which will impact the combustion chemistry and potentially the net amount of NOx generated (or reduced) in the calciner. Per Mr. Greer, ‘the use of riser duct firing at Lyons has the potential to lower the NOx emission factor because the kiln flue gas passes through an oxygen-deficient, fuel rich

zone in the riser duct, where free radicals react with and eliminate a portion of the NO_x generated in the rotary kiln.’”

Hofmann, p. 5-16 to 5-17.

Response

Mr. Hofmann implies that riser duct firing at Lyons, in combination with oxygen injection, reduced NO_x emissions during the period of interest: 1997 - 2000. At Lyons, riser duct firing currently occurs in the vertical ductwork that connects the feed end of the rotary kiln to the calciner. The practice consists of burning a relatively small portion of the fuel in the riser duct under oxygen-deficient conditions. Riser duct firing was *not* part of the pyroprocess during the period of 1997-2000. The documentation I have seen is not clear as to the intended purpose of riser duct firing at Lyons but riser duct firing at the Lyons Plant was being studied around June 6, 2001.⁶⁵ Riser duct firing at Lyons appears to have commenced on or about October 23, 2001, a date well after the period of interest of my expert report.⁶⁶

8. ANNULUS BOOST FAN

CEMEX Contentions

“The Annulus Boost fan project occurred nearly three years after the projects primarily associated the Lyons Expansion Project. The new Annulus Boost fan improved the combustion characteristics in the kiln leading to product quality improvement and potentially a modest increase in clinker production. The timing and nature of the Annulus Boost fan project[,] however, does not satisfy the US EPA’s 1989 and 1993 aggregation policy guidance with respect to the earlier projects.”

Hofmann, p. 5-22.

“Although a similar project was originally contemplated to be included within the scope of the LEP, it appears that the facility abandoned the project in 1998. It further appears that the installation of the fan in 2000 was an independent decision and distinct from the LEP. For these reasons, it is my opinion that this project need not be aggregated for PSD purposes with other projects undertaken at the facility.”

Hofmann, p. 5-30.

“An annulus boost fan was installed in 2000 to provide a higher tip velocity flame. Unstable flames in a rotary kiln—which were occurring at the Lyons plant—adversely affect kiln operations and are unsafe for plant equipment and personnel. An unstable flame generally has a variable ignition point or plume

⁶⁵ Memorandum from Mike McCabe, Re: Lyons Site Visit, June 6, 2001, CMXLYO00402655.

⁶⁶ October Highlights, CMXLYO00043829.

length and during kiln upset conditions can result in the flame going out, with an explosion resulting a few seconds later. Additional concerns with a burner pipe with a low tip velocity (*e.g.*, less than 7000 fpm) is the possibility of the flame propagating back into the burner pipe and into the coal mill system resulting in a fire or potential explosion in the fuel preparation system. Lyons addressed these concerns through installation of an annulus boost fan. As a result, improved flame stability was achieved[,] which improves kiln operational stability, which will in turn typically reduce NOx emissions.”

Young, Summary of Opinions, Opinion No. 6, p. 7.

“Thus, by improving the thermal efficiency and stability of the kiln system, the overall NOx emissions from the Lyons facility were likely reduced by the addition of the annulus fan while some production gains may have been realized.”

Young, p. 45.

Response

In addition to the reasoning in my expert report, as explained below, I believe Mr. Hofmann is incorrect when he states that the annulus boost fan should not be considered as part of the changes made at Lyons beginning in 1997. As explained further below, I disagree with Young’s conclusion that the booster fan likely reduced NOx emissions.

A. CEMEX’s Addition of the Annulus Boost Fan Was Part of Its Lyons Expansion Plan

CEMEX’s Contentions

“Hofmann contends that the annulus boost fan project should not be aggregated together with the other changes CEMEX beginning in 1997 made at the Lyons facility. Without support from or reference to the record, he contends that the project initially contemplated by CEMEX in late 1997 as part of its expansion planning was canceled and that the ‘the installation of the fan in 2000 was an independent decision and distinct from the LEP.’”

Hofmann, p. 5-30.

Response

I disagree. As early as December 1997, CEMEX planned to install a coal booster fan at the Lyons Facility.⁶⁷ In its expansion plan, CEMEX provided:

⁶⁷ Lyons Plant Expansion – Phase I, Project Scope – Revised December 15, 1997, CEMEX_NEIC_00003299 at 3304.

“Coal Booster Fan: The coal tip velocity of the kiln burner is low compared to optimum levels for a direct fired burner. The result is a long and lazy flame which produces a long burning zone. This proposal will add a high static booster fan to the coal circuit which will increase the tip velocity to the optimum level.”⁶⁸

CEMEX’s intention to go forward with this project is confirmed in a document titled: “Lyons Plant Expansion, January 1998.”⁶⁹ Installation of the annulus boost fan was delayed in 1998 to allow Lyons to determine the correct burner pipe momentum, placement, and alignment.⁷⁰ Further, funding for the project was “carried over” in the 1999 plant budget,⁷¹ and the 2000 plant budget.⁷² The project was implemented in the summer of 2000.⁷³ Because the annulus boost fan project was implemented within the presumptive 5-year period following the modification of the calciner, and as CEMEX’s own budget planning documents indicate that the project was carried forward from conception in 1997 to execution in 2000, I believe it proper to consider the annulus boost fan with those other projects that occurred during this period.

B. The Annulus Boost Fan May Have Contributed to Increased NOx Emissions

In my expert report, I note that the increased temperature of the shorter, more intense flame and the higher volume of primary air resulting from the annulus boost fan would have tended to increase NOx generation in the burning zone of the rotary kiln.⁷⁴ Mr. Young feels that the annulus boost fan would have decreased NOx generation because of improved kiln stability. Although anything that improves kiln stability has the potential to decrease NOx emissions, for the reasons presented here, I believe Mr. Young is likely to be wrong in this instance. In an October 2000 email from Kirk Light, Lyons Plant, to Mike McCabe, CEMEX Combustion Group, Light observes that “the new [annulus] burner has definitely created a more intense and consiste [sic] flame which has resulted in a smaller burning zone.”⁷⁵ A “smaller burning zone” is consistent with a “more intense” flame, i.e., higher temperature flame, which, in turn should result in the increased generation of thermal NOx. Light’s observation supports my projection about the flame shape and temperature.

In addition, Dr. Kevin Davis discusses flame characteristics and fuel NOx generation in his expert report.

“Low NOx burners rely upon slowing the mixing of fuel and air and creating fuel-rich regions in the volatile flame. This can result in a balancing act between flame stability and controlled mixing. If the flame is not well attached and stable,

⁶⁸ *Ibid.*

⁶⁹ Southdown, Inc., Lyons Plant Expansion, January 1998, CEMEX_NEIC_00003452 (“In addition, a booster fan will be added to the kiln burner to improve flame characteristics.”).

⁷⁰ Text and Figures for FCT AFE, CMXLYO00045595.

⁷¹ Lyons 1999 Capital Budget, Item 15, Coal Mill Booster Fan (Carry Over), August 28, 1998, CMXLYO00155783 at 155800.

⁷² Lyons 2000 Capital Budget, Item 14, Coal Mill Booster Fan (Carry Over), January 14, 2000, CMXLYO00415808 at 415814.

⁷³ See, Memorandum from John Lohr to Steve Bryan, Re: Lyons Kiln Status, August 17, 2000, CMXLYO00157315.

⁷⁴ Greer, p. 36.

⁷⁵ Email from Kirk Light to Mike McCabe, Re: Clinker, October 17, 2000, CMXLYO00402668 – 2669.

then there is significant mixing of the combustion air surrounding the coal stream with the coal stream itself prior to the release of volatile material. This creates air-rich conditions that result in greater conversion of the coal nitrogen to NO_x.⁷⁶

When equipped with an annulus boost fan, the Lyons multi-channel burner pipe appears to be the antithesis of the low-NO_x burner Dr. Davis describes. The annulus fan increases the amount of high-velocity primary air available to the flame that in turn causes the flame to detach from the burner to produce the air-rich, stable conditions that could promote the formation of fuel NO_x. In support of himself, Dr. Davis subsequently states, “[i]n practical coal combustion applications, NO_x emissions will typically increase significantly when excess oxygen is increased from 1-3%.⁷⁷ The annulus boost fan does increase available primary air at the combustion site.

In his memo, which exclusively addresses the annulus fan, Mr. Light further observes, “[t]here has been an increase in NO_x, but I need a couple more days [sic] better quantify what this increase has been.⁷⁸ Mr. Light’s observation certainly supports my conclusion that the annulus fan would serve to increase the potential for thermal NO_x generation. As presented above, Dr. Davis explains how the potential for fuel NO_x generation also may have increased. I have seen no evidence that would cause me to conclude that the benefits of improved kiln operating stability advocated by Mr. Young would outweigh the inherent increase in NO_x generation potential that accompanies the use of an annulus boost fan in support of the multi-channel kiln burner.

9. CKD RECYCLING

CEMEX’s Contentions

“Since the mid 1990s, the Lyons plant has increased cement kiln dust (CKD) recycling (instead of wasting it, the plant utilizes it in the kiln as an improved raw mix). CKD is a calcined material and requires minimal (if any) additional fuel in the pyroprocessing system. Increased recycling of CKD would increase clinker production and reduce NO_x emissions in terms of per ton of clinker. To the extent increased CKD use resulted in higher clinker production levels and any associated NO_x increases, such increases should be excluded from any PSD analysis because of use of alternate raw material exclusions provided under the PSD regulations. Review of CKD clinker production between 1995 – 2006 time period shows that clinker production levels have remained almost flat. This, along with the other factors affecting NO_x variability, makes any attempt to attribute specific production increases to any given project very difficult. Finally, the recently proposed portland cement Maximum Achievable Control Technology (PC MACT) standards recognize benefits of CKD recycling.”

⁷⁶ Davis, p. 10.

⁷⁷ *Id.* at p. 16.

⁷⁸ Email from Kirk Light to Mike McCabe, Re: Clinker, October 17, 2000, CMXLYO00402668 – 2669.

Hofmann, Summary of Opinions, Opinion No. 20, p. 1-6 and 1-7.

“The Lyons plant recycles CKD. Adjusting the total clinker production numbers with CKD free clinker production, it appears that clinker production almost remained constant during the alleged Lyons Expansion Project period.”

Hofmann, p. 5-28.

“In general, production capacity of a cement kiln is determined by subtracting CKD removed from the total kiln feed or alternatively from the produced clinker.”

Hofmann, p. 5-28.

“Increased CKD recycling would beneficially impact NO_x emissions of the system (*i.e.*, lower NO_x emissions per ton of clinker produced).”

Hofmann, p. 5-30.

“By recycling CKD back to the pyroprocessing system, the overall energy efficiency of the system improves.”

Young, p. 27.

Response

It does not appear that Mr. Hofmann fully understands the cement pyroprocess and the nature of the materials involved – particularly CKD. As a previous cement chemist, I doubt that many current or former cement chemists would regard CKD as “improved raw mix.” It is partially calcined raw mix that most often carries the unwanted chemical burdens of higher percentages of sodium, potassium, sulfur, mercury, semi-volatile metals (*e.g.*, lead) and sometimes chlorine. Physically, CKD is hot, caustic, fluid, and difficult to handle.

The *production rate* of a cement kiln in any given time period is determined by subtracting from the weight of the raw mix used in that period the corresponding weight of the “loss on ignition” of the raw mix and the corresponding weight of CKD not recycled to the process, if any, corrected to a loss free basis. For convenience, a factor is usually developed by which raw mix is converted into equivalent clinker by simple multiplication. If the “loss on ignition” changes or the amount of recycled CKD changes, the factor must change if the clinker production rate is to be stated accurately.

As I explain in further detail below:

1. Clinker production is clinker production and it can’t be “adjusted” in the manner presented by Mr. Hofmann. CKD is not clinker and it must be processed into clinker. The clinkering process and the product are the same no matter whether

virgin raw mix or a mixture of virgin raw mix and recycled CKD are used as the feed to the pyroprocessing system.

2. Because of the differences in chemical composition, CKD cannot be treated the same as clinker in mathematical operations.
3. The material collected in the Kiln and Hart baghouses is not technically CKD and should not be considered as such in analyses advocating the advantages of recycling CKD.
4. The use of recycled CKD in clinker production should require less thermal energy and generate lower NO_x emissions than when an equivalent amount of virgin raw mix is used but that is only part of the energy and environmental considerations regarding CKD recycling at the Lyons plant.
5. Even though recycled CKD requires less energy to be converted into clinker than virgin raw mix requires, a substantial amount of thermal energy is still required to complete the calcination step and to form the clinker minerals.

A. The Recycling of Cement Kiln Dust in Clinker Production Is a Common Practice

CKD is a byproduct of cement manufacture. As its name suggests, it is a finely-divided, solid material that is captured in particulate matter control devices. It may be recycled to the process from whence it came, disposed on the land, or used for a variety of beneficial purposes. CKD may be generated as a necessary consequence of cement clinker production or it may be purposefully generated to remove sodium and potassium salts from the process and hence the clinker. The cement manufacturer has money invested in partially processed CKD so it is in the manufacturer's financial interest to convert as much CKD into cement clinker as possible. Disposal of CKD on the land is expensive and carries the potential for long-term liability.

B. "CKD Free" Is a New Concept to the Cement Industry and Requires Clarification

In almost all dry-process cement kiln systems, including the one at Lyons, when CKD is recycled, it is mixed or included with virgin raw mix prior to its reintroduction into the pyroprocess. Under these circumstances, it is impossible for an individual clinker nodule or a batch of clinker to be "CKD free," *i.e.*, not contain solids that result from the recycling of CKD. The term "CKD free" clinker is not included in the normal cement industry vocabulary. As used in Mr. Hofmann's expert report, the term "CKD free" clinker should not be misunderstood as describing clinker that does not contain CKD. As presented in Table 5-8 of his report, "CKD free" clinker appears intended to be clinker that could have been produced from virgin raw mix had CKD not been recycled.

C. The Chemistry of In-Process Materials – Kiln Feed, CKD and Clinker – Are Similar But Not Exactly the Same

Within any relatively brief time period, the kiln feed, the CKD, and the clinker in the same kiln system are very similar in chemical composition except for one primary parameter, “loss on ignition.” This is because all these in-process materials originate from the same raw materials but have experienced different degrees of thermal treatment.

“Loss on ignition” is an analytical term that represents the loss in weight of a sample of material after it is heated to a high temperature, e.g., 1750°F, for a period of time sufficient for the sample to achieve constant weight. All of the volatile constituents or those constituents that can be evolved as a gas at that temperature will have been liberated from the sample. When cement kiln feed is heated, the constituents that are liberated during the test (and also during clinker production) primarily include carbon dioxide, and free and chemically combined water. To a lesser extent, hydrocarbons and highly volatile salts and metals also may be vaporized. Because CKD has already experienced some thermal treatment in the kiln, the constituent that evolves under the high temperature conditions of the test (and also during clinker production) is primarily residual carbon dioxide and, to a much lesser extent, volatile salts. Heat-treated cement clinker does not have constituents that can be liberated during the test and is considered to be “loss free.”

The examples in the following table illustrate the differences in “loss on ignition” for in-process materials at the Lyons Plant. In this illustrative example, I have assumed the operation of the alkali bypass (ABP) and the return of CKD at the relative rates shown. As shown, there is a wide range of values for “loss on ignition” of the in-process materials. During normal operation of the Lyons pyroprocess, the values are not constant. When explaining or attempting to understand the Lyons pyroprocess, “loss on ignition” must be considered.

Material	Percent Calcined	Loss On Ignition
Virgin Raw Mix	0	36
Kiln Feed including 10% CKD Return	7.70	33.23
Calcliner Product	71.0	10.40
Material Collected in Kiln and Hart Baghouses	71.0	10.40
CKD Returned to Process (90% Alkali Bypass + 10% Kiln and Hart)	77.0	8.28
Alkali Bypass CKD	77.7	8.03
Clinker	100.0	0.00

D. The Entry in Table 5-8 for “Cement Kiln Dust Return” in 1996 Did Not Include Nine Months of Data

From CEMEX records in the data base, it was determined that the total amount of CKD returned to the pyroprocess in 1996 was 37,271 tons rather than 11,838 tons, thereby making the correct entry in Table 5-8 for “CKD Free Clinker Production” 393,001 tons. When this number is properly corrected, it becomes clear that the 1997 increase in clinker production is the result of something other than the return of CKD.

E. The Mathematics in Mr. Hofmann’s Table 5-8 Is Incorrect

Based on the illustrative values for “loss on ignition” shown in the table above, it is clear that the mathematics in Mr. Hofmann’s Table 5-8 is analogous to subtracting apples from oranges. It is simply incorrect to subtract the annual quantity of recycled CKD that contains a residue of carbon dioxide from the “loss free” total annual clinker production to obtain “CKD free” clinker – whatever that quantity is intended to represent. The CKD must be converted mathematically to a “loss free” or a “clinker equivalent” basis before it is subtracted from total annual clinker production to obtain the equivalent amount of clinker that theoretically was produced from virgin raw mix. To make the calculation on an annual basis, the average “loss on ignition” of the CKD must be known to correct the weight of the returned CKD to its clinker equivalent. The establishment of an average annual value for “loss on ignition” of the CKD that is returned to the Lyons pyroprocess is a daunting task because of the different operating scenarios for the ABP and the periodic manufacture of low-alkali clinker during which all or most of the CKD is discarded. At a minimum, the testing of daily samples of the returned CKD would be required to achieve a minimal level of accuracy. Those data have not been located in the database. My estimated values for “loss on ignition” will have to suffice for the following illustration.

According to Table 5-8 of the Hofmann expert report, the Lyons plant recycled 68,371 tons of CKD to the kiln in 2000. Using my illustrative value of 8.28% “loss on ignition” for CKD returned to the process, this quantity of CKD would have been equivalent to 62,710 tons of clinker.⁷⁹ The 62,710 tons of clinker resulting from CKD recycling would not be in addition to or less than the reported 539,992 tons of annual clinker production in 2000. *It was simply included in the total annual production of clinker.*

F. Knowing the Source of the Cement Kiln Dust, i.e., the Combination of the Kiln and Hart Baghouses or the Alkali Bypass Baghouse Alone, Is Important to Understanding the Lyons Pyroprocess and the Potential Effect of Cement Kiln Dust Generation and Recycling on NOx Emissions

Absent any information or data to the contrary, it must be assumed that the CKD tabulated as “Cement Kiln Dust Return” in Mr. Hofmann’s Table 5-8 includes CKD collected in all three baghouses serving the pyroprocess. Nowhere is it stated in the Hofmann expert report how much of the CKD quantified in Table 5-8 originates from the Kiln and Hart baghouses or

⁷⁹ $[68,371 \text{ tons} \times (1 - 0.0826)] = 62,710 \text{ tons}$

how much originates from the ABP baghouse. The Process Flow Diagram for the Lyons Plant, last revised October 23, 1994, shows multiple pathways for the CKD that is collected in all three kiln system baghouses to find its way to the kiln feed stream.⁸⁰

At Lyons, the material collected in the Kiln and Hart baghouses simply is the partially calcined product of the calciner that is entrained in the flue gases leaving the preheater cyclone and the conditioning tower. This material is essentially identical to the feed to the rotary kiln that leaves the preheater by a different path. It should be returned to the pyroprocess on a continuous basis for economic considerations alone. This is the practice at Lyons where it is returned in the kiln feed. It would be an unusual circumstance (*e.g.*, a mechanical breakdown) if this material was removed from the process, even temporarily. Using the term “CKD” to identify a material that is simply undergoing an internal process cycle is technically incorrect. *Nevertheless, it is a fairly common practice in the cement industry to describe this and similar materials as CKD.* If the weight of Kiln and Hart baghouse material is included in the numbers in the “Cement Kiln Dust Return” column of Figure 5-8, it would be appropriate to subtract that weight from the total currently shown in that column because it is technically *not* CKD.

Only the material collected in the ABP baghouse is the true CKD at Lyons. Depending on the needs of the plant, all or part of this CKD may be removed from or recycled to the pyroprocess. Only when the entire volume of ABP material is *not* being returned to the pyroprocess is clinker being produced that did not originate with CKD as part of the kiln feed.

If partially calcined CKD is recycled to the process, less thermal energy will be required in the pyroprocess to produce a ton of clinker from the CKD than from the virgin raw mix it replaces. When recycled CKD is used as part of the kiln feed, the potential clinker production under most scenarios likely would be increased up to the limits of the ID fan. Therefore, the recycling of CKD does not necessarily mean that mass NO_x emissions will decrease because more or less the same amount of fuel will be burned to produce a measurable but only slightly increased quantity of clinker. In point of fact, the NO_x emissions following a change in CKD usage may change slightly, up or down, or remain the same depending on many other factors unrelated to CKD recycling. The fact remains, in a fan limited pyroprocess, normal practice is to burn the maximum amount of fuel allowed by gas flow through the pyroprocess and then adjust the clinker production rate to coincide with the available thermal energy. All other things being equal, the amount of mass NO_x emissions is heavily dependent on the amount of fuel being burned.

G. The Recycling of Cement Kiln Dust from the Alkali Bypass also Carries Penalties in Fuel Consumption and NO_x Generation As Well As the Benefits in Fuel Consumption and NO_x Emissions Suggested by Mr. Hofmann

At the Lyons plant, the operation of the ABP system carries with it economic and environmental penalties associated with the thermal losses from the pyroprocessing system. When it is necessary to produce low-alkali clinker, these penalties cannot be avoided. To remove volatile sodium and potassium from the pyroprocess, very hot gas containing these elements is removed from the hood at the feed end of the rotary kiln. The thermal energy in this

⁸⁰ Process Flow Diagram for the Lyons Plant, last revised October 23, 1994, CMXLYO00038055.

gas is lost to the system because the gas is exhausted to atmosphere through the stack after being mixed with and chilled by ambient air, and dedusted in the ABP baghouse. Were it not for use of the ABP, this gas would have gone to the calciner where its inherent thermal energy would have been partially recovered. To avoid a drop in clinker production when the ABP is in use, this energy must be replaced in the kiln and/or calciner by additional fuel burning that always carries a concurrent increase in NO_x generation and emission potential. In addition, NO_x generated in the rotary kiln passes directly through the ABP to the atmosphere without the opportunity for its elimination by reaction with free radicals in the riser duct or calciner.

H. Under Certain Circumstances, the Alkali Bypass System at Lyons Can Be Used to Supplement the Induced Draft Fan and Thereby Increase the Clinker Production Potential of the Fan-Limited Pyroprocess

If the Lyons plant is willing to accept the inherent thermal inefficiency and resulting economic penalty, the ABP system can be used to increase the effective ID fan capacity and increase clinker production in an otherwise fan-limited system. These penalties can be avoided by not increasing clinker production when the ABP is not needed to produce low-alkali clinker. To accomplish this objective, the chemistry of the raw mix must be changed to avoid excessive alkali input or high-alkali clinker must be accepted as a satisfactory product. The CKD collected in the ABP dust collector must be returned to the process in a manner similar to the return of the material collected in the Kiln and Hart baghouses. Under these circumstances, the return of this material to the process is not CKD recycling. In effect, the ID fan and the ABP fan are operated in parallel thereby increasing overall flue gas flow through the pyroprocess that results in increased potential for fuel burning and clinker production. The potential for additional NO_x generation and emissions would track an increase in fuel burning and the bypassing of the NO_x-containing flue gas from the rotary kiln around the riser duct and calciner.

An examination and comparison of *Operating Reports* and *Daily Production Reports* for the Lyons plant the years of 1997 and 1998 reveal many instances when the alkali bypass fan was running but low-alkali clinker was not being produced. This is a definitive indication that the process just described was being used. I will leave it for others to decide if this is a qualifying change in the method of operation of the Lyons pyroprocess that also should have resulted in the consideration of PSD applicability.

APPENDIX A

References Directly Used in the Preparation of the Rebuttal Report

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Deposition of Stephen Goodrich, December 9, 2010.

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